Maximilian Dittgen

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Education

Cornell University — Ithaca, NY
M.Eng. in Computer Science — *Expected Graduation May 2026*

Cornell University — Ithaca, NY
B.S. in Computer Science, Minor in Asian American Studies, 3.97/4.00 GPA, Dean's List — *Aug. 2021 – May 2025*
Teaching Assistant: CS 4300 (Language & Information), CS 4852 (Market Design / Game Theory)

Experience

Amazon Web Services (AWS) | **C, Linux** — Berlin, Germany
Software Development Intern, EC2 Kernel/Hypervisor — *September 2025 – December 2025*
- Designed and implemented hardware-accelerated interrupt injection mechanisms for the EC2 Nitro Hypervisor on ARM, improving I/O performance for networking-intensive workloads by 5–10%
- Developed kernel APIs, KVM ioctls, and QEMU commands for interrupt mechanism control and testing at runtime
- Authored and submitted a multi-patch RFC series to the upstream Linux kernel, including documentation, benchmarks, and validation via Linux KVM selftests
- Identified and contributed three auxiliary bug fixes to Linux KVM selftests, all merged into mainline Linux

Capital One Travel | **TypeScript** — New York, NY
Software Engineering Intern — *May 2025 – August 2025*
- Implemented an automated ETL service to aggregate hotel, flight, and entertainment offerings from multiple internal databases and expose a unified API for third-party vendors
- Built AWS Glue job to ingest and process 2 million offerings daily, integrated with a Fastify API for external access
- Deployed resilient AWS infrastructure (S3, EC2, IAM) to support a scalable and fault-tolerant ETL pipeline

Capital One | **Java** — McLean, VA
Software Engineering Intern — *May 2024 – August 2024*
- Developed automated data pipeline to capture and publish Zelle payment events to Snowflake data warehouse, handling 150,000 production requests per day
- Designed aspect-oriented pointcuts in Java Spring to splice data publication flows into existing Zelle API endpoints
- Collaborated with fraud analyst end users to understand service requirements and ensure data scalability

Exostellar | **C, Linux, Go** — New York, NY
Software Engineering Intern — *May 2023 – August 2023*
- Optimized Xen hypervisor to speed live migration of Docker containers between AWS EC2 cloud instances by 50%
- Created Linux kernel driver to dynamically adjust memory allocated to a container during migration, enabling more workloads to run a single host virtual machine
- Implemented gRPC service to manage resources and processes of cloud containers from local user interface

Projects and Publications

Constructive Comment | JavaScript — November 2025
- Published and presented as Shahid, F., **Dittgen, M.**, Naaman, M., Vashistha, A., Examining Human-AI Collaboration for Co-Writing Constructive Comments Online, Proc. ACM Hum.-Comput. Interact. (CSCW), 2025
- Developed UI and UX of Generative AI writing aid that coaches rural, low-resource users in India to draft constructive Reddit posts and comments, working under Cornell's Computing for Global Development Lab

Saliency-Guided Burst Denoising for Low-Light Photography | C++ — May 2025
- Created image processing pipeline and UX for denoising low-light images similar to iPhone's Night Mode
- Implemented C++ denoising pipeline using optimized corner detection, homography alignment, bilateral filtering, and saliency-guided burst image fusion, enabling selective freezing or motion blur for moving objects

Foret | Python, Vertex AI | Y Combinator Best Hack Award at HackMIT 2023 — September 2023
- Investment analytics API that scrapes, summarizes, and fact-checks corporate sustainability reports into short blurbs
- Leveraged LangChain and Vertex AI Text-Bison to summarize long-form ESG report PDFs
- Trained sentiment model to fact-check corporate sustainability reports against web-scraped news articles

Technical Skills

Languages: Java, C/C++, OCaml, Python, MATLAB, R, Go, HTML/CSS, Bash Script
Tools and Frameworks: Git, Docker, AWS Serverless/EC2, Linux Kernel, Jenkins, Pytorch/Numpy, Spring, IntelliJ IDEA